Speakers

Arqum Ahmad, CEO, Antrax Labs and AI Scientist at Leonardo, Italy (Military Aircraft Division)

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in an unexpected situation? "Simulators like Carla from Intel may not cover all possible scenarios in real world and only give a jump start for analysing certain scenarios from real world. There is a larger cosmos outside simulations which needs to be taken care of, often called one-of or edge cases due to non-standard driving, system failure or other environmental factors," informs Naresh Neelakantan.

2. Reinforcement learning. After loading a 3D map of the city in a simulator, the next step is to assign certain vehicles with poor driving skills. But there are perhaps more ways to badly drive a vehicle than favourably. It is unknown to what extent a bad driver can be simulated. Therefore, it is required to simulate bad driving in the real world and collect live data for better analysis. And with this, deep learning methods can pretty much pick up the complexities of a real-world traffic scenario and other factors such as light intensity, weather, stray animals, road conditions, and other vehicles present on Indian roadways. Once the risks and accidents are modelled, the traffic can be stabilised to a large degree.

Regarding deep learning applications, more datasets result in better system accuracy. Some of the famous datasets include:

Astyx HiRes. It is a combination of LiDAR, radar, and 3D object detection.

UC Berkeley DeepDrive. It has 100,000 video sequences with diverse annotations for image-level tagging, object bounding boxes, and full-frame segmentations.

Google Landscape. Wide dataset of different landscapes; ideal for training a driverless car in different landscapes with varying weathers.

"This will reach a tipping point towards six or nine sigma kinds of accuracy when more Turing Machine-like AI methods coupled with quantum computing could provide deeper insights from now on," says Naresh Neelakantan.

Where is intelligence heading?

As pointed out by Dr Rahul Kala, autonomous technology has a role in providing a safe transportation experience. For instance, a lack of awareness among truck drivers on the limit to which a person can continuously drive a vehicle causes fatigue, resulting in accidents. This is where technology comes in to prevent fatal situations.

Emphasising the role of academicians, he says academic institutes are quite active in publishing data and actively discuss ideas for solving real-world problems. "Startups, academia, and government can collaborate on projects for the future of both EVs and autonomous vehicles. In India, there are hundreds of problems—let's solve them one-by-one," says Dr Kala.

Integration of intelligent fast tracks into autonomous driving systems allows connected AV training for safer and faster travel. Also, deployment of fast tags facilitates quicker movement near toll plazas.

Government policies

Government agencies and test labs are coming up with new strategies to test EVs. The FAME 2 policy with regards to accelerating the adoption of EVs, which was approved in 2019 with hundred billion rupees of investment, is one such example. Thanks to such an initiative, 40,000+ EVs are currently running on Indian roads and more are expected in the years to come. Besides, 5G connectivity is also expected to play a huge role. On top of that, the government of India is already pushing forward *Make in India* and *Atmanirbhar Bharat* schemes to give Indian industry a boost.

To minimise the infrastructure bottleneck of EV charging, the Indian government, as a part of the FAME 2 policy, has planned to deploy charging stations across petrol pumps nationwide so that these are readily available for use.

Regarding policies on autonomous vehicles, there seems to be a slight hesitation in implementing them as the government is pretty much concerned about increased job losses on account of vehicle autonomy. Therefore, to be on a safe side, it is more supportive for EVs (partly due to environmental issues as well). Taking into consideration several other factors—such as Indian roads, potholes, weather, etc—it is a long way to go for autonomous vehicles to enter Indian markets.

In summary, new EV standards continue to be developed, which can be leveraged for promoting wider adoption. When it comes to intelligent vehicles, a reliable structure involving edge computing, deep learning, and IoT should be present for enhanced judgement in unforeseen situations. Finally, the government has introduced many new policies that can be leveraged for wider adoption of EVs. Despite the reluctance to adopt autonomous vehicles, India would one day inevitably have a balanced ecosystem in which both humans and autonomous vehicles co-exist. EFY

This article is based on the panel discussion 'Challenges faced in testing cars of the future (connected, intelligent, EV, etc) and possible solutions' during December edition of India Technology Week 2020. The panel was chaired by Naresh Neelakantan, senior architect - System Software at Sasken Technologies